# Evolving Native-Noise Controllers for Safe Completion with Frozen Manipulation Policies

Frozen Jar comparison and separately audited model-assisted preparation

Sachin Mohan
University of the Witwatersrand
Johannesburg, South Africa
2699183@students.wits.ac.za

## ABSTRACT

Can a compact controller recover safer task completion from a frozen generative manipulation policy under a small interaction budget? We implement an observation- and instruction-conditioned adapter that adds a bounded four-dimensional offset to native sampling noise. Its 176 parameters use measured robot state, executed action history and frozen image/text features; the family-specific VLA and native execution contract remain fixed. A bounded 320-call decoder audit on ManiGuard Jar and Clutter reproduces native sampling on all 32 same-noise identity pairs and confirms influence on the executed action prefix in all four tested directions per family. Median perturbed-prefix arm differences are .00319 and .00187 rad, respectively. Four full development episodes establish integration and timing, without a matched estimate of physical benefit. We freeze a 60-valid-episode allocation: four development episodes, eight shared initial episodes, sixteen descendant episodes each for evolution and random search, and sixteen independent checks of native, fixed, evolved and random-search controllers. All eight shared initial episodes end unfinished; the highest-utility constant never grasps. In the first generation, each method has zero completions and four violations in eight episodes. Two training episodes close the lid safely but never grasp or transport the jar. The best EA utility exceeds random search through nonviolating unfinished execution, including one uncontacted case. Independent results remain pending. A separate 48-call diagnostic finds gripper-coordinate influence at a stronger noise offset, with arm coupling and no physical benefit tested. A separate CPU model-assisted prototype's recorded-data audit fails prediction and calibration gates; a small second physical experiment is proposed rather than reported as completed. Native-noise steering and evolutionary VLA steering have precedents. The contribution sought is an explained safe-completion benefit per total interaction, with competent learning comparisons, replication and transfer still required.

## CCS CONCEPTS

• **Computing methodologies**;

## KEYWORDS

evolutionary robotics, frozen policies, latent steering, safe manipulation

*Research draft, 20 October 2026,*
2026.

## 1 INTRODUCTION

Completing a manipulation task and respecting its physical specification are different outcomes. A robot may transport a jar with its lid open, or avoid every violation while never finishing. ManiGuard makes these distinctions measurable through task checks and passive runtime monitors [5]. We investigate whether a small controller can improve safe requested completion around its family-specific frozen vision-language-action policies.

The adapter steers the generator's native noise before decoding an action chunk, conditioning on current observations, the actual instruction and causal executed action history. Evolution optimizes a small parameter matrix from complete physical simulator outcomes. The VLA, feature encoder and native execution contract remain fixed. This physical comparison deploys no learned safety filter, world model or online action veto. Section 7 separately presents implemented CPU preparation, its failed factual audit and the proposed second experiment; those components do not alter the running comparison.

Earlier restraint and feedback pilots [3] motivate this change without validating it. The earlier full-strength finalist achieved 4/8 safe completions with two violations, against identity's 5/8 with none. An adaptive query trigger never refreshed early. A saved-trace diagnostic verified a direction-preserving cap but found few trigger opportunities. These are historical results from different controllers and a legacy rendering profile, not controls for the present experiment.

We separate four questions: does adaptation help native execution; does conditioning help a fixed bias; does evolutionary retention and mutation help relative to other adaptation methods; and do fitted weights transfer? Our completed evidence establishes integration, decoder influence and unsuccessful first-generation physical adaptation. A matched Jar comparison is in progress under a protocol frozen before comparative feedback. Its small sample can provide effect-size leads, not establish optimizer superiority or cross-family transfer. Shared initialization now supplies eight valid full episodes without safe completion on its two cases. First-generation results add partial lid closure without completed grasp or transport.

## 2 RELATED WORK AND CONTRIBUTION BOUNDARY

DSRL learns in the latent-noise space of a frozen diffusion policy, including generalist-policy adaptation [10]. Flow Reversal Steering maps coarse reference actions through a reversed flow and learns auxiliary noise policies; it also studies repeated or averaged

noise across chunk positions [8]. Principal Steering Subspaces constructs a low-dimensional basis from decoder responses and controls its leading directions with SAC [4]. Noise-space adaptation, chunk repetition and low-dimensional steering are therefore not novel by themselves. Our present basis is random and fixed, not response-informed.

Evolutionary VLA steering also has precedents. VLA-Pilot evolves action proposals under an inferred steering objective during inference [2]. Online-ES uses executed trajectory perturbations and feedback-weighted training to update model parameters [11]. Here the base VLA stays frozen while selection updates a reusable conditional adapter using specification-labelled episode outcomes. This defines an empirical question, not a priority claim. A conventional EA or optimizer substitution alone is insufficient novelty.

The intended contribution is a reproducible, mechanistically explained improvement in safe requested completion per total interaction. Evidence must distinguish useful steering from a tuned constant, and evolutionary advantage from random search and competent ordinary learning. One fitted adapter working with several family checkpoints is the longer-term target. Reusing its interface with per-family refitting is a weaker claim than transferring the same weights.

# 3 PHYSICAL EVALUATION BOUNDARY

At query time $\tau$, the frozen family policy decodes native Gaussian noise $\epsilon_\tau \in \mathbb{R}^{16\times 32}$ from two RGB views, robot input and the prompt. Native output transforms produce physical joint-target/gripper commands. The first eight are executed unless native termination occurs sooner. The adapter changes noise rather than action horizon.

Let $V$ be raw native `ltl_violated`, without engagement gating. Requested completion $C_{\text{req}}$ requires native success and the pinned family checklist. For Jar, closure must be observed strictly before lifting, remain true at every off-support observation and at the end, and lifting must occur. The final native held/intersection goal must also hold. This stricter sequence check is reported beside the unmodified benchmark verdict. Safe completion and training utility are

$$S_{\text{req}} = C_{\text{req}} \wedge \neg V, \qquad U = \mathbf{1}(S_{\text{req}}) - 2\mathbf{1}(V). \tag{1}$$

A violating completion scores $-2$; unfinished nonviolating execution scores zero. This is a declared penalized objective. Reporting retains all four joint completion/safety categories, raw violations and denominators. Contact/grasp indicators contextualize unfinished episodes; fewer violations can reflect inactivity.

The monitor supplies evaluation labels and never changes commands. Monitor predicates, object poses, task/family/checkpoint identifiers and goal state are excluded from the adapter. Timeouts are valid outcomes, not infrastructure failures, and do not alone identify their physical cause.

# 4 NATIVE-NOISE CONTROLLER

## 4.1 Causal features

The feature vector $\phi_\tau \in [-1,1]^{44}$ comprises a bias, seven arm positions divided by $\pi$, two measured finger positions divided by .04 m, seven previous executed environment-input arm targets minus current arm positions divided by .2 rad, and three end-effector displacements since the previous query divided by .05 m. Robot/history coordinates are clipped. These targets precede downstream native-controller clipping; they are neither achieved joint positions nor final motor commands. Initially the previous target equals measured joints and displacement is zero.

The remaining 24 coordinates are eight each from overview RGB, wrist RGB and the actual instruction. Frozen CLIP ViT-B/32 [7] produces 512-dimensional embeddings. Each embedding is normalized to unit Euclidean norm. Fixed Gaussian-QR projections, seeded 20261017, produce eight coordinates, multiplied by $\sqrt{512}$, clipped to $[-5,5]$ and divided by five. No feature fitting uses study outcomes. Images are encoded at every native query and text once per episode, including in the native reference. The conditioning path is implemented; its physical value is unestablished. A recorded-input audit finds only two distinct training instructions and centered text-block rank one across shared initialization. Text connectivity therefore does not establish semantic generalization rather than case-specific offsets.

## 4.2 Bounded steering with native stochasticity

For genome $W \in [-1,1]^{4\times 44}$, define $c_\tau = \tanh(W\phi_\tau)$ and

$$z_{\tau,h} = \epsilon_{\tau,h} + .35Bc_\tau, \qquad h = 1,\dots,16. \tag{2}$$

The first seven rows of $B \in \mathbb{R}^{32\times 4}$ are a reduced QR basis of a Gaussian $7\times 4$ matrix with PCG64 seed 20261019. Remaining rows are zero. The offset repeats across the chunk and has Euclidean norm at most $.35\sqrt{4} = .70$ per position in noise coordinates. This bounds sampling input, not joint motion or violation probability. Gripper/padding noise are not directly steered, but decoder coupling can affect their outputs. No persistence, learned noise scale or activation gate is used.

For $W = 0$, an explicit branch calls unchanged native `Policy.infer` without an explicit-noise argument. Nonzero controllers preserve the native RNG split and Gaussian draw before adding the offset. A zero noise tensor would define a different policy. Receipts retain RNG transitions, noise/observation hashes, offsets, decoded physical chunks and source hashes. The runtime logs steered proposals and final environment inputs. It does not buy an additional unsteered chunk at every query; same-noise comparisons are confined to the bounded decoder audit.

## 4.3 Matched search

Shared initialization evaluates identity, positive and negative constants, and a uniform random conditional matrix on two cases. Constants set only $W_{0,0} = \pm\operatorname{atanh}(.5)$, yielding $\pm.175$ along the first basis direction. The random matrix uses seed 20261019. Both methods receive these same eight episodes, counted once globally and attributed to both information budgets. The better constant becomes the fixed reference, with ties selecting positive. This tunes two declared constants, not an extensive fixed-policy baseline.

Each method evaluates two generations of four candidates on two common cases per generation. Both freshly re-evaluate the previous cohort's best candidate. EA adds three children from alternating top-two parents using iid Gaussian weight mutation with standard deviation .25 and clipping to $[-1,1]$. Random search

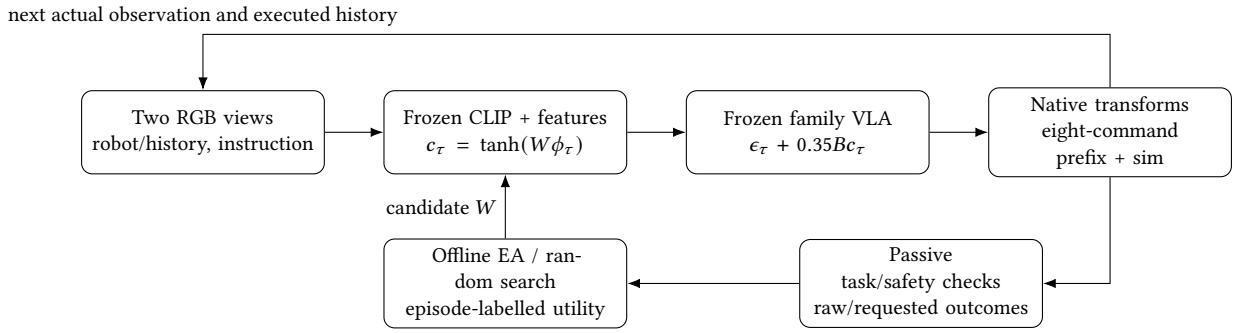



**Figure 1: Implemented information boundary. Candidate weights are fixed within an episode; the native Gaussian draw remains present. Outcome labels guide selection of later candidates without vetoing execution. All conditions pay the same feature extraction cost. The VLA also receives its unchanged native observation inputs. No learned safety filter or world model is deployed.**

**Algorithm 1** Frozen comparison after four development episodes

**Require:** Pinned policy, features, domain, cases, seeds and budget
Evaluate four shared controllers on two cases (8 episodes)
Select fixed reference from the two constants
Initialize EA and random search with the same scored data
**for** generation $g = 1, 2$ **do**
    Propose both four-candidate cohorts, including incumbents
    **for** candidate slot $j = 0, 1, 2, 3$ **do**
        **for** method in the frozen alternating slot order **do**
            Evaluate its candidate on two common cases
            Verify both physical receipts and record scores
        **end for**
    **end for**
    Rank each complete cohort for subsequent selection
    Preserve all attempts, ancestry, RNG and physical records
**end for**
Seal native, fixed and the two final-cohort finalists
Check four conditions on four blocks (16 episodes)
Report outcomes, block differences, activity and costs

instead draws three uniform matrices from the same domain, with the same retained-incumbent cost. Ranking uses summed utility and stable identifiers for ties. Only the last complete cohort supplies each finalist, preventing best-of-repeat promotion. Both method RNGs use seed 20261020. Descendant execution interleaves methods at each candidate slot, using EA/random at even slots and random/EA at odd slots. Lineage and pending/completed evaluations are persisted.

# 5 PROSPECTIVELY FROZEN PHYSICAL PROTOCOL

## 5.1 Scope and native execution

Jar and Clutter provide development exposure; Cabinet and Dusty remain physically untouched by this study. Jar/Stack/Lid/Clutter have programme-level history. Jar is the primary pilot and Clutter the declared capacity fallback. The selected protocol was frozen at 04:27:23 UTC on 20 October before comparative feedback, using development timing and the authority criterion. It selects Jar. This pilot does not train or test one shared cross-family adapter.

Execution uses native 640 × 640 DLAA rendering, two actual camera views, the verified extended-finger Franka, assisted grasping, 20 Hz policy control, 60 Hz rendering and 120 Hz physics. Hard horizons are 2,000 controls for Jar and 1,100 for Clutter. Preprocessing, gripper binarization, native clipping and termination are unchanged. Jar uses checkpoint step 7400, revision `1d84eda07031`; Clutter uses step 7100, `f572e33bdb49`. Full revisions and file hashes remain in study records.

Jar training tasks are 0019/0009, with base seeds $6101910 + 10g + i$, generation $g = 0, 1, 2$ and task index $i = 0, 1$. Checking tasks 0016/0012/0011/0003 use seeds $6101940 + i$. These blocks are reserved from this optimizer's feedback. Historical validation allocation does not establish absence of earlier programme exposure; we conservatively describe them as exposed tasks with independent checking. The original main-study test partition remains excluded. Native per-scene sampling-seed derivation is retained.

## 5.2 Allocation and hardware balancing

Table 1 specifies the 60 planned valid full episodes. At most four genuine infrastructure failures may be replaced, yielding a 64-full-attempt ceiling. Every partial/invalid attempt and discarded candidate is charged. Valid timeouts and adverse scores cannot be replaced. Separate engineering used six starts: three failed during preparation and three reached 32 controls each. The 96-control engineering allowance is exhausted. Physics before callback attachment is unknown, not zero. The original 320-call decoder allowance is exhausted. A separately authorized 48-call follow-up, reported below, changes neither this physical allocation nor the controller used in the frozen comparison.

Two workers each colocate simulation and policy on one RTX 3090. Training case $i$ uses worker $(i + g) \bmod 2$ for both methods. Denoting native, fixed, EA and random by $N, F, E, R$, checking orders are $(N, F, E, R)$, $(F, R, N, E)$, $(E, N, R, F)$ and $(R, E, F, N)$. Adjacent pairs run concurrently; positions zero and two use worker zero. Every condition therefore runs twice per GPU. One admission ledger enforces the global attempt ceiling. Process identities

**Table 1: Frozen full-episode allocation. Shared initialization is acquired once. The sixteen-episode independent check is protected from adaptation and tuning.**

| Stage | Valid episodes |
|---|---|
| Development, Jar + Clutter | 4 |
| Shared initialization: four controllers, two cases | 8 |
| EA: two generations, four candidates, two cases | 16 |
| Random search: same structure and domain | 16 |
| Independent check: four conditions, four blocks | 16 |
| Planned total | 60 |
| Maximum starts with infrastructure replacements | 64 |

and terminal receipts permit reattachment without duplicate physical launches.

Capacity protects 56 post-development episodes plus four replacement slots as 30 two-worker waves. Its per-wave bound is the maximum of 180 s, 1.25 times the slowest complete development process, and a startup-plus-native-horizon throughput estimate with the same margin. Including 900 s for lifecycle/preservation, Jar requires 15.008 h. The original window was insufficient; the user subsequently authorized the full comparison on this instance. The fixed cutoff is 19:30 UTC and independent local stop 20:00 UTC on 20 October. Horizons/rendering were not reduced to fit.

### 5.3 Analysis and uncertainty

The final comparison will retain all four requested completion and safety categories and raw benchmark completion per condition, with paired differences by checking block. Initial robot/image hashes are checked: common seeds do not prove identical camera observations. Contact/grasp indicators contextualize noncompletion. Cost curves use common cumulative physical control counts and retain all adaptation, failures and discarded candidates. Model calls, encoder work, elapsed time and allocated GPU-hours are separate counters. Shared episodes count once physically and once within each method's information budget.

Setup, decoder calls, idle allocation and preservation are charged. GPU-hour and running-cost estimates are upper bounds over elapsed wall time from the original provider start, counting both GPUs. The own-instance API does not expose verified billed runtime; a container restart is not credited as unbilled time. Four checking episodes per condition and one optimizer seed provide descriptive effect sizes, not a superiority or safety guarantee. Zero violations in $n$ independent episodes implies an approximate 95% upper rate bound of $3/n$, not zero; four episodes are particularly weak evidence.

## 6 COMPLETED PHYSICAL AND DECODER EVIDENCE

The first-generation summary is dated 20 October 09:45 UTC. A subsequent elite recheck is explicitly dated below. Independent checking remains unopened.

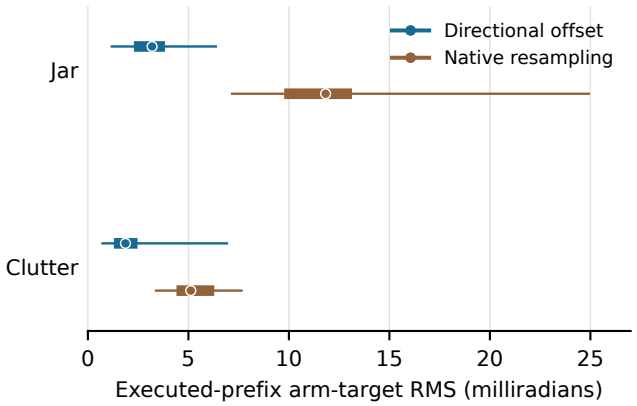



**Figure 2: Arm-target RMS differences over the executed eight-command prefix. Thick intervals show quartiles, thin intervals the range and circles the median. There are 128 directional comparisons and eight native-resampling comparisons per family, clustered within eight saved observations. These are decoder responses, not independent physical trials or safety measurements.**

### 6.1 Identity and decoder influence

Eight saved observations per family, from two development tasks, each use two native noise draws. Each draw receives two same-noise baselines and positive/negative offsets along four directions: ten calls and 320 total. All sixteen identity pairs per family pass the declared $10^{-5}$ absolute/relative tolerance. Full remote recovery also reproduces noise hashes and RNG transitions without new model calls.

Perturbed-prefix arm RMS ranges from .001131 to .006422 rad for Jar, median .003193, and .000666 to .006975 rad for Clutter, median .001868 (Figure 2). Native-resampling medians are .011827/.005118 rad. All four directions per family pass the frozen criterion: observation-level mean RMS at least .001 rad on at least half the observations, with two passing directions required. The smallest individual Clutter response need not exceed this observation-level threshold. No executed-prefix gripper flips occur among 128 directional probes per family. Coupling elsewhere is not excluded.

A 32-control Clutter integration check uses an untrained dense conditional matrix, exercising all 176 weights including 96 semantic weights. Removing semantic features offline changes its code by RMS .3781 on factual observations. This verifies a software path, not semantic competence or a physical counterfactual.

### 6.2 Full development outcomes

Table 2 preserves four full episodes on different tasks/seeds. Their native/bias differences do not estimate an adaptation effect. Jar native reaches the raw benchmark goal but fails the closed-lid sequence: requested unsafe unfinished, versus raw unsafe completion. Jar bias and Clutter native reach their full horizons, both valid outcomes rather than replaceable failures.

Development closeout contains ten starts, 4,394 controls, 4,464 environment steps and 26,987 known physics ticks. There are 551

**Table 2: Full development episodes. $C$ is requested completion, $V$ raw violation, $A$ controls and $Q$ policy calls. Each row uses a different task/seed.**

| Family/controller | Task | $A$ | $Q$ | $C$ | $V$ | $U$ |
|---|---|---|---|---|---|---|
| Jar native | 0013 | 666 | 84 | 0 | 1 | −2 |
| Jar positive bias | 0017 | 2000 | 250 | 0 | 0 | 0 |
| Clutter native | 0039 | 1100 | 138 | 0 | 1 | −2 |
| Clutter negative bias | 0010 | 532 | 67 | 1 | 0 | 1 |

**Table 3: Shared training initialization, two episodes per controller. SC/UC are safe/unsafe requested completion; SU/UU are safe/unsafe unfinished episodes. $V$ counts violations and $G$ episodes with a recorded grasp. Random here is the shared untrained conditional initializer, not a selected random-search finalist.**

| Controller | SC | UC | SU | UU | $V$ | $G$ |
|---|---|---|---|---|---|---|
| Native | 0 | 0 | 1 | 1 | 1 | 1 |
| Positive constant | 0 | 0 | 1 | 1 | 1 | 0 |
| Negative constant | 0 | 0 | 2 | 0 | 0 | 0 |
| Random conditional | 0 | 0 | 0 | 2 | 2 | 0 |

physical-policy calls and 320 decoder calls. Encoding records 1,102 images, seven text calls, 23.519 s extraction and 38.443 s loading; physical-policy request wall time is 293.776 s. These totals precede the running comparison and are not final study costs. Callback-unobserved initialization physics remains unknown.

Clutter native has a 574.6 s cold startup versus its partner's 99.9 s. Logs identify a locked shared Kit cache and shader compilation. The episode then completes normally and retains its cost in capacity selection. This technical delay is distinct from its valid adverse physical outcome.

## 6.3 Completed shared initialization

All eight shared initialization episodes reach the full 2,000-control horizon without requested or raw benchmark completion (Table 3). Each controller receives the same two nominal task/seed cases, costing 4,000 controls and 500 policy calls. Shared acquisition therefore costs 16,000 controls and 2,000 policy calls globally, attributed once within each optimizer's information budget.

The negative constant's utility sum zero leads native and positive at −2 and the random initializer at −4. It never grasps and contacts the target in only one of its two episodes. This lead concerns nonviolating unfinished execution, not improved safe completion. The frozen rule selects it as the fixed reference and initial incumbent of both searches. Native task 0019 goes off support while open at action 537; closure first becomes true at 638. Positive task 0009 instead loses uprightness at 1116 while still supported, with no recorded grasp. These are factual failure descriptions, not counterfactual explanations of a steering effect.

The initial robot input matches native in all six comparisons of a different controller on the same case. Both RGB views differ in every pair. Stored native-noise hashes and RNG transitions agree at all 1,500 paired query indices across those six streams. The common random stream is realized while observations remain different. These are correlated consistency checks over two cases, not independent outcome trials.

**Table 4: Supplemental gripper-noise influence. Entries count saved observations with at least one binarized gripper-command flip in the first eight decoded actions under either sign. The last column is the largest arm-target difference across those positions and the ±2.0 probes. These are correlated decoder inputs, not physical outcome trials.**

| Family | Magnitude 0.5 | Magnitude 2.0 | Max arm (rad) |
|---|---|---|---|
| Jar | 0/4 | 3/4 | .05875 |
| Clutter | 0/4 | 4/4 | .01773 |

## 6.4 Exploratory gripper-coordinate diagnostic

Four generation-one incumbent rechecks subsequently evaluate the same negative constant on the new training seed panel. Each method scores $[0, -2]$: no requested or raw completion and one violation per pair. At this diagnostic cutoff, all twelve comparison episodes are unfinished, six violate, and only one ever grasps. Eight never command gripper closure; the other four each show actual finger closure. None of their arm proposals changes at the outer environment bounds. These observations did not identify a unique failure cause. Offspring evaluation was still pending at that earlier diagnostic boundary; its completed first generation is reported below.

The frozen basis excludes direct support on the gripper-noise coordinate. During a diagnostic pause, a separately frozen 48-call follow-up tests coordinate seven (zero-based) on eight saved middle/last development frames, four per family from two episodes. Each input receives a native draw, an explicit same-noise repeat and offsets ±0.5 and ±2.0 on that coordinate at all sixteen chunk positions. No simulator or encoder runs. This raises total decoder requests to 368 while retaining the original audit and physical cohort unchanged. All eight additional same-noise identities pass at the original $10^{-5}$ tolerance.

The stronger offset changes gripper commands in both families (Table 4), with accompanying arm changes. Its positive sign opens the gripper in all four Clutter frames, including two factual held-target frames; its negative sign closes it in the two unfinished Jar-bias frames. Conditional timing and physical benefit remain untested. Three inputs already lie on violating factual trajectories, and no Jar input demonstrates a successful closure transition. A flip there cannot establish prospective violation prevention. This is influence in a candidate new representation, with no repair applied to the frozen physical controller.

An independent offline sensitivity analysis uses 3,334 factual Jar query rows and 205 Clutter rows from completed development/training episodes. The three persisted EA mutations change their parent codes by RMS .5410/.5825/.4130 on the Jar pool; only .990%/1.110%/.127% of their code components have magnitude at least .95. Inactive mutation or general tanh saturation is therefore unsupported as the current explanation. These are noise-space

**Table 5: First-generation training results on two fixed cases per candidate. Entries give the two utilities, raw violation count $V$, and safe supported lid- closure episode count $L$. All rows have zero requested/raw completion and grasp. Incumbents are paid re-evaluations, not new genotypes.**

| Method | Candidate | Utilities | $V$ | $L$ |
|---|---|---|---|---|
| EA | Incumbent | $[0, -2]$ | 1 | 0 |
| EA | Child 1 | $[-2, 0]$ | 1 | 0 |
| EA | Child 2 | $[-2, -2]$ | 2 | 0 |
| EA | Child 3 | $[0, 0]$ | 0 | 1 |
| Random | Incumbent | $[0, -2]$ | 1 | 0 |
| Random | Draw 1 | $[0, -2]$ | 1 | 0 |
| Random | Draw 2 | $[0, -2]$ | 1 | 1 |
| Random | Draw 3 | $[0, -2]$ | 1 | 0 |

sensitivities on correlated factual inputs, with longer trajectories weighted more. They do not measure decoded actions, semantic competence or physical counterfactual benefit.

## 6.5 Completed first generation and ordered progress

All sixteen first-generation episodes reach 2,000 controls and 250 policy calls. Neither method records requested or raw completion or a grasp. Each has four violating and four nonviolating unfinished episodes, seven contact episodes and population utility sum −8 (Table 5).

The last EA child wins the cohort with utility zero, versus random search's best −2. Its first case closes the lid on support at observation 1,187 but never grasps; its second never contacts the jar. This is a training-utility lead through nonviolating unfinished execution, not a safe-completion benefit. Random draw 2 closes safely at observation 482 and remains supported and closed through 2,000, but likewise never grasps. It ties all random candidates under the frozen utility; stable identifier ordering correctly retains the incumbent rather than rewarding this undeclared milestone.

Across the eight shared initial and sixteen first-generation episodes, none completes, twelve violate and only the initial native episode ever grasps. The twelve safe unfinished episodes comprise ten without an ordered milestone and two with supported lid closure only. Nine comparison episodes issue closing gripper commands, but none after safe supported closure. These factual complementary partial behaviours motivate a separately declared continuation test; they do not establish composability or uniquely diagnose failure. Milestone rewards, tie-rule changes and transfer of a random-search controller into EA would change the method and are excluded from this comparison.

The selected EA controller is re-evaluated on the next frozen seed panel before mutation. By 10:05 UTC, both generation-two elite cases again score zero, with no completion, grasp or lid closure; one never contacts the jar. The closure milestone therefore has not repeated on those two seeds. Other generation-two candidates and all sixteen protected checking episodes remain pending. Finalists will be sealed from the last complete cohorts before checking; no optimizer superiority or transfer benefit is claimed.

Development, engineering, shared initialization and completed generation one cost 52,394 policy controls, 6,551 physical-policy requests and 317,123 known physics ticks. Initialization physics before callback attachment is unknown. The separately charged 368 decoder requests bring those completed-phase VLA requests to 6,919; these are not final study totals and exclude subsequent generation-two physical work. Recorded-data CPU model costs remain separate below.

# 7 MODEL-ASSISTED FOLLOW-UP AND SECOND EXPERIMENT

**Status: completed CPU preparation and audit; physical follow-up unexecuted.** This separate study asks whether progress-aware control and qualified short-horizon prediction can improve task continuation without paying for a broad physical search. Its CPU implementation does not change the frozen comparison above.

## 7.1 Implemented model and control interfaces

A five-member episode-bootstrap ensemble predicts the next 44 compact features from the current and previous features, a history-valid flag and eight known executed eight-dimensional commands. Its 153 inputs feed ridge regressors with 32 fixed nonlinear random features. Separate heads estimate a new violation within the prefix and five task predicates: closed lid, holding, goal intersection, uprightness and support. A readout evaluated on actual future features separates perception/readout error from imagined-dynamics error. This compact transition prototype does not generate future RGB observations or simulate a complete closed-loop VLA episode. A prediction about a decoded prefix is a surrogate, not measured fitness.

A prospective controller augments the 44 causal inputs with five learned beliefs. Its $5 \times 49$ weights produce four arm-noise coefficients and one gripper-noise coefficient, with coefficient persistence. Prototype settings are arm scale .35, gripper scale 2.0 and persistence .8; they are not physically validated magnitudes. Native Gaussian noise is retained. Privileged simulator predicates cannot silently replace the unqualified learned beliefs. A separate bounded joint-target residual provides a prospective representation control; its source-matched CPU clipping wrapper still requires actual runtime limits and physical identity validation.

The action-changing filter accepts a supported low-risk prefix, substitutes an eligible alternative, or abstains. Proposed, filtered and actually executed commands remain distinct. Candidate allocation scores the surviving prefix and shares predictions for identical filtered prefixes at the same factual anchor. Absent qualification, allocation preserves abstention and cannot authorize execution. Decoder requests for discarded proposals remain charged.

The ROSARL interface implements its practical reward/value-extrema penalty rule, requiring a declared external value estimator and one penalty frozen before candidate comparison [9]. It is not a trained task policy or a complete ROSARL baseline. The shortest-path theory with unsafe absorbing outcomes does not certify this finite-horizon partially observed system. SafeDreamer already combines world-model planning and safety objectives [1]; this combination alone is not the proposed contribution.

**Table 6: Separate CPU model audit on one diagnostic episode. Values are feature RMSE; rows are correlated measurements within that episode. The predeclared gate requires a model/persistence ratio at most .90 in every block. All three fail.**

| Feature block | Model | Persistence | Ratio |
|---|---|---|---|
| Robot/history | .16805 | .08516 | 1.973 |
| Overview | .04381 | .02848 | 1.538 |
| Wrist | .06380 | .05481 | 1.164 |

## 7.2 Completed factual audit and failed gate

The audit protocol was frozen before fitting. Eight shared-initialization episodes on tasks 0019/0009 supply 1,992 fitting transitions; one development episode on task 0017 supplies 249 calibration transitions, and one on task 0013 supplies 83 diagnostic transitions. Splits are by whole task, all previously exposed. Independent checking outcomes are excluded. Final prefixes without a following encoded observation are omitted, and risk fitting excludes prefixes starting after a latched violation. These ten existing episodes cost 18,666 original acquisition controls, retained as reused information.

Actual-feature readout MAE is .191 for closure, .209 for holding, .662 for uprightness and .210 for support, above the .10 gate. The fitting set contains positive closure and holding in only one episode each, no in-goal positive episode, and no safe completion. Its observed closure follows a violation. Calibration has one episode against the minimum 20. The conservative fallback therefore rejects every prefix; its 100% negative-prefix alarm rate is not the false-alarm rate of a qualified classifier. Counterfactual-action coverage would still require justification after adequate factual calibration.

The audit consumes 27,412 training-row uses and 5,395 model-head row queries, with zero new physical episodes, simulator controls, VLA calls or encoder images. Sixteen focused CPU tests validate the initial assembled interfaces; later allocation/residual and clipping-wrapper suites add 12 and 15 checks, respectively. These checks establish software behaviour, not manipulation success. The failed prediction/coverage gate excludes this model from physical action selection.

## 7.3 Proposed small continuation screen

Before further model-guided optimization, test whether a known closure behaviour can hand control back for body grasping and transport. Freeze one controller chosen using exposed training data, its prior-information charge, task, fresh seed, execution order and every switching parameter before the first outcome. Use four references on one development case: native only, closure only, a fixed clock handoff, and a privileged supported-lid-closure handoff. The privileged reference is an engineering feasibility control, with additional actor information; it cannot establish equal-information learned-method superiority.

The proposed trigger requires three consecutive post-control observations with the jar closed and supported and no preceding violation. Initial observation zero does not count. Losing closure/support before the next eight-control query cancels qualification. Switching occurs at that query and remains latched to native execution, without resetting its Gaussian RNG. A new trial resets the phase history; partial failures are charged and cannot be resumed as valid full episodes. Source pins and per-query branch/RNG receipts need a new validity reader; the original constant-genotype verifier is insufficient. This contract has CPU checks, not physical RNG-continuity or continuation evidence.

The first screen is capped at four full starts, including partial and failed starts. If all are valid, it evaluates the four references once each. Expand by a separately frozen stage on a second case only if a prespecified safe ordered body-grasp or completion criterion passes. A final native-versus-selected-nonprivileged comparison may use four additional full episodes on fresh seeds: at most 12 valid episodes and two charged infrastructure replacements, 14 starts total. A valid timeout or adverse outcome is not replaced. Four full-horizon starts require at most 8,000 policy controls and 1,000 native policy queries, including partial/invalid work within that screen. Initialization, settling, encoders, partial failures, allocated time and preservation require their own numerical ceilings in the execution-owned protocol. No extra decoder sweep is needed for this handoff screen. The original study's protected sixteen checking episodes remain separate.

This is a prospective reduction of the earlier 16-valid-plus-two-replacement capability proposal, not an additional allowance. Execution must freeze the complete staged protocol and verify resource/deadline capacity before launch. Even a successful small screen cannot supply the minimum 20 independent calibration episodes for the learned filter. Failure calls for diagnosis rather than a larger optimizer sweep and does not establish irreversibility.

## 7.4 Scientific comparison after capability and model repair

A later funded comparison must separate model-free and model-assisted evolution, matched random allocation/search, ordinary task-policy learning with the same filter, and native, filtered-native and strong fixed references. Give competitors the same observations, training labels, prior data, model access and total interaction budget. Progress-aware training must retain requested safe completion as the outcome and remove progress fitness after a violation. Filter, ROSARL and evolutionary retention each need their own ablation. Replicated optimizers, independent confirmation and frozen-controller family transfer remain necessary before asserting the intended GECCO contribution.

# 8 LIMITATIONS AND REMAINING EVIDENCE

This interface can consume several family checkpoints, but the pilot fits only within Jar. A random four-direction basis may miss useful behaviour. A bounded noise offset need not imply safe contact dynamics. The 176-parameter domain is large relative to two-case candidate feedback. Noisy selection, adverse behaviour and identity finalists are valid outcomes. If a constant explains an improvement, conditioning is unestablished; if random search matches EA, evolutionary advantage is unestablished. An EA lead over uniform

random search combines parental reuse with a local Gaussian proposal distribution. Attributing it to evolutionary reuse additionally needs a fixed-parent Gaussian-search ablation and a matched local-search comparison under separate, prospectively funded evaluations.

An episodic parameter-REINFORCE interface over the same bounded matrix is implemented and CPU-checked, including interrupted-batch recovery. It has zero physical evaluations and is outside the frozen minimum allocation. It does not replace a competent state-conditioned DSRL-style latent SAC baseline. That baseline, matched tuning and independent optimizer replication are required before claiming practical evolutionary superiority. A direct-action residual baseline is additionally needed to attribute any advantage to the latent representation.

The cross-family target requires freezing one fitted controller and checking new tasks and an adaptation-held-out family with its own frozen base policy. Response-informed bases, persistence, sampling-distribution control and internal activation gates remain prospective, with no physical result here. A proposed 156-episode adaptation/transfer bridge is separately budgeted and unexecuted. Neither extensions nor tuning may consume the final checking allowance.

## 9 REPRODUCIBILITY AND PRESERVATION

The launch revision is `4783369c5b59`. Upstream pins are ManiGuard `6bf4c61a1f16`, BEHAVIOR `88454bd04f75` and openpi `215abfb217db` [6]. Separate uv-managed environments serve simulation (Python 3.10, Torch 2.6/CUDA 12.4), policy (Python 3.11, JAX .5.3) and CPU tooling. Comparative receipts retain actual GPU UUIDs, source hashes, causal features, decoded chunks, executed inputs and raw/requested outcome evidence. An independent reader verifies physical records before assigning fitness.

Code, compact evidence and frozen protocols are preserved in https://github.com/Sachin2911/Safety-Dial. The protocol is `configs/maniguard/latent-evolution/capacity-20261020-v2.json`; its study README documents reconstruction. Separate private Hugging Face model/dataset repositories store immutable stage archives with remote revision, privacy, size and SHA-256 verification. Remote restoration re-verifies all 28 complete episodes through generation one, the next two elite cases, all 32 original identity pairs and eight supplemental pairs without new physics or model calls. Raw runs, licensed assets, credentials and base-policy weights are excluded from public Git.

## 10 CONCLUSION

Native-noise steering is implemented with conditional inputs and verified decoder influence in two family policies. Shared initialization yields no safe-completion gain; its highest-utility controller remains unfinished without grasping. Both first-generation populations have zero completion and four violations in eight episodes. Two safe lid-closure cases reveal partial progress, with no subsequent grasp or transport; the EA selection lead includes nonengagement. The continuing frozen Jar comparison tests native execution, a tuned constant, evolution and matched random search while protecting independent checking and retaining unsuccessful candidates. Its result must determine the strength of the contribution, including a null or adverse finding. Ordinary learning, replication and transfer remain necessary for the broader claim. The separate model-assisted prototype fails its factual qualification gate. Its staged continuation experiment must establish measured task progress before larger model-guided search or action-changing filtering is justified.

## ACKNOWLEDGMENTS

This BSc Honours research at the University of the Witwatersrand is supervised by Geraud Nangue Tasse. AI coding assistance supported implementation, diagnosis and drafting; recorded empirical evidence and its stated limits govern the claims.

## REFERENCES

[1] Weidong Huang, Jiaming Ji, Chunhe Xia, Borong Zhang, and Yaodong Yang. 2024. SafeDreamer: Safe Reinforcement Learning with World Models. https://doi.org/10.48550/arXiv.2307.07176 arXiv:2307.07176 ICLR 2024.

[2] Zhuo Li, Junjia Liu, Zhipeng Dong, Tao Teng, Quentin Rouxel, Darwin Caldwell, and Fei Chen. 2026. Towards Deploying VLA without Fine-Tuning: Plug-and-Play Inference-Time VLA Policy Steering via Embodied Evolutionary Diffusion. https://doi.org/10.48550/arXiv.2511.14178 arXiv:2511.14178

[3] Sachin Mohan. 2026. Intervention-Aware Execution Adaptation around Frozen Manipulation Policies. https://github.com/Sachin2911/Safety-Dial/blob/4783369c5b599580e26f9c3a2723ee55518c46a3/docs/research/reports/maniguard-phase3-part1-gecco/draft.pdf Internal supervisor discussion draft, 19 October 2026; historical execution and saved-trace results.

[4] Jialeng Ni, Nathan Zhao, and Kunpeng Song. 2026. Principal Steering Subspaces for Online Adaptation of Frozen Generative Robot Policies. https://doi.org/10.48550/arXiv.2609.33765 arXiv:2609.33765

[5] Yiyan Peng, Philip Wang, Simon Sinong Zhan, Yiqi Lyu, Zhenyang Ni, Jixin Yan, Fiorelli Wong, Ruochen Jiao, Hang Yin, Xinyu Cao, Huajie Shao, Manling Li, Ruohan Zhang, and Qi Zhu. 2026. MANIGUARD: A Benchmark and Data Suite for Specification-Grounded Safety Evaluation and Improvement of Robotic Manipulation. https://doi.org/10.48550/arXiv.2608.17386 arXiv:2608.17386 [cs.RO]

[6] Physical Intelligence. 2026. openpi: Pinned policy implementation. https://github.com/Physical-Intelligence/openpi/tree/215abfb217dbac7d5f1273282331b9b1866c0479 Revision 215abfb217dbac7d5f1273282331b9b1866c0479; accessed 18 October 2026.

[7] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. https://doi.org/10.48550/arXiv.2103.00020 arXiv:2103.00020

[8] Andy Tang, William Chen, Andrew Wagenmaker, Chelsea Finn, and Sergey Levine. 2026. Improving Robotic Generalist Policies via Flow Reversal Steering. https://doi.org/10.48550/arXiv.2606.13675 arXiv:2606.13675

[9] Geraud Nangue Tasse, Tamlin Love, Mark Nemecek, Steven James, and Benjamin Rosman. 2023. ROSARL: Reward-Only Safe Reinforcement Learning. https://doi.org/10.48550/arXiv.2306.00035 arXiv:2306.00035

[10] Andrew Wagenmaker, Mitsuhiko Nakamoto, Yunchu Zhang, Seohong Park, Waleed Yagoub, Anusha Nagabandi, Abhishek Gupta, and Sergey Levine. 2025. Steering Your Diffusion Policy with Latent Space Reinforcement Learning. https://doi.org/10.48550/arXiv.2506.15799 arXiv:2506.15799

[11] Gongxin Yao, Yongsheng Zhao, Jiayin Deng, Deng Liang, Han Gao, Lei Zhao, and Baoping Cheng. 2026. Online Evolution Strategy for Flow-Matching VLA Policies via Self-Supervised Trajectory Distribution Optimization. https://doi.org/10.48550/arXiv.2609.38855 arXiv:2609.38855